

Standard AI project management workflow for the integration of Industrial AI within an organization

AI projects are categorised into two segments based on the users:

- **Vertical:** targets the Employees(The services providers)
 - Goal: Improve and Enhance the process adopted by the employees, Resulting to a better accuracy and efficiency, Achieving Scalability
- **Horizontal:** Targeting the Customers
 - Goal: Enhance user experience, Raise Engagement Resulting to a significant Improvement of the final product

We call this type of AI projects: **Industrial AI**

- The perspective we adopt within Amzil AI developed from the conjunction of our expertise in Tech, Business and Product management
- The purpose is to understand and acquire the methods needed to organize the industrial development of an AI project

Standard AI product Development Framework

1. **Feasibility study:** Determine the cost and viability of a project.
2. **Proof of concept:** Test The ability of a technical solution to meet the need in a controlled, limited environment.
3. **Pilot:** Production testing limited in time and space (The main goal would be to collect user feedback)
4. Runs stably in production, scalable



● Feasibility study:

A. Project Objectives and Strategic Alignment

- Clear definition of objectives: Specify what the project aims to achieve and how it fits into the broader strategic goals of the organization.
- Alignment with business strategy: Assess how the AI project supports the overall objectives and strategies of the business.

B. Technological Assessment

- Available technologies: Examine existing AI technologies and their suitability for the project's needs.
- Required innovations: Identify areas where additional technological developments or innovations are necessary to meet the project's objectives.
- Technological maturity analysis: Evaluate whether the necessary technology is mature enough to be successfully implemented in the project.

C. Data Analysis

- Data availability: Assess the availability of data needed to train and test AI models.
- Quality and relevance of data: Examine the quality, relevance, and representativeness of the available data.
- Ethical and legal considerations: Consider regulations regarding privacy protection, data ownership, and ethical use.

D. Skills and Resources Evaluation

- Internal skills: Identify available internal skills and expertise and additional training needs.
- Recruitment needs: Determine if external skills need to be acquired.
- Cost and resources estimation: Estimate the financial, material, and human resources required for the project.

E. Market Analysis

- Market demand: Assess the existence and size of the demand for the proposed solution.
- Competition: Analyze existing competitive solutions and their performance in the market.
- Commercial viability: Evaluate the commercial viability of the project, including potential revenue models.

F. Risks and Compliance

- Risk identification: Identify technical, financial, operational, and reputational risks associated with the project.
- Mitigation measures: Propose strategies to mitigate these risks.
- Regulatory compliance: Ensure compliance with relevant laws and regulations, especially concerning AI and data use.

G. Preliminary Project Plan

- Planning and milestones: Define a project plan with key stages and milestones.
- Technology roadmap: Develop a roadmap for the necessary technological development for the project.



● **Proof of Concept:**

A. Selection of Representative Data

- Choose a dataset that accurately reflects real-world usage conditions.
- Ensure the quality, relevance, and diversity of the data to effectively test the AI algorithm.

B. Minimalist Design and Development

- Develop a simplified version of the solution to test the idea without excessive time and resource investment.
- Use agile tools and technologies for rapid iteration.

C. Technical and Functional Validation

- Verify that the AI model functions as intended under controlled conditions.
- Test the solution for specific use cases to assess its performance and applicability.

D. Performance Evaluation

- Employ robust evaluation techniques to measure the algorithm's effectiveness against defined objectives.

● **Scalability Study:**

1. User Scalability

The AI system can serve an increasing number of users or clients without degradation in user experience or system responsiveness.

2. Data Scalability

The ability to maintain or improve performance as the data increases.

3. Operational Scalability

The AI product can be expanded or upgraded with minimal effort or disruption to existing operations. This includes being able to deploy updates, add new features, or integrate with additional systems and technologies easily.

4. Infrastructure Scalability

The hardware and software infrastructure can support growth. This might involve moving to more powerful servers, utilizing cloud computing resources more effectively, or optimizing software to run more efficiently on existing hardware.

5. Cost Scalability

The costs associated with scaling up (including computational resources, storage, and operational costs) should be proportional and manageable relative to the growth and the value generated by the AI product.

Road to product : phases matrix

	Study	POC	Pilot	Product
Goal	Document (report, planning, pricing)	Software (internal)	Software (client)	Software (client)
Completion time	1 week to 3 months	1-6 months	3 months to 1 year	6 months to 2 years
Operating time	N/A	1 week to 3 months	3 months to 1 year	Unlimited
Stability	N/A	Low	Average	High
Scalability	N/A	None	Low	High
Users	N/A	1-6 (dev team)	1-100	1-∞
Cost	XS	M	L	XXL

Study - Classification of surgical tools.

Market segment

Healthcare

Field application

Classification

Context and issues

Training operators to sort and reassemble surgical instrument kits after sterilization is a lengthy process. There are thousands of references, and the difference between two instruments is sometimes very subtle, leading to errors when reassembling kits.

Study

To address this issue, XXXX scientific teams carried out a study to assess the most appropriate technical approach to this complex problem, due to the working environment and the multiplicity of references to be recognized.

Results

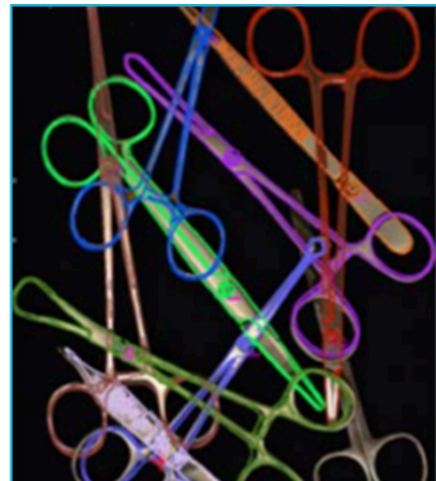
- > Precise definition of the algorithmic work to be carried out
- > Detailed dimensioning of the developments to be carried out
- Knowledge of the state of the art
- Breakdown of development batches to produce the customer product

Deliverable

Complete study to define the most appropriate technical approach for the following lots

Integration

N/A



POC - Deep Turnaround.

Market segment

Aeronautic

Field application

Security

Context and issues

The control processes on aircraft stations involve many service providers and are subject to rigorous planning. Potential errors during these processes can lead to security breaches and consequent financial losses for airport groups.

Solution déployée

XXXX has developed a computer vision software solution that enables real-time analysis of events to make the aircraft operational again for take-off. Thanks to these real-time detections, the application enables the identification of unforeseen events such as delays in the schedules that have been started. The information is then reported via the ADP computer system and a forecast of the new flight schedule can be set up.

Return on investment

> Increased efficiency in detecting risk situations

> Improved feedback to operators

● Improved safety system reliability



Deliverable

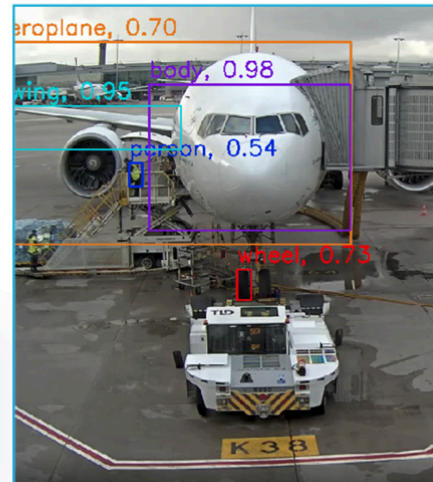
Software capable of analyzing video streams and producing a detailed schedule of detected events, as well as a delay forecast.

Intégration

Flux SAFIR

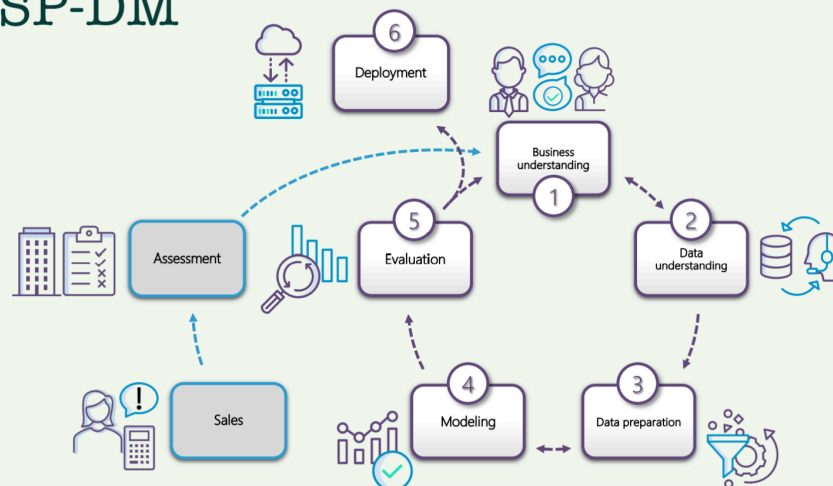
Environment

Local / On-premises



● From Proof of concept to production:

CRISP-DM



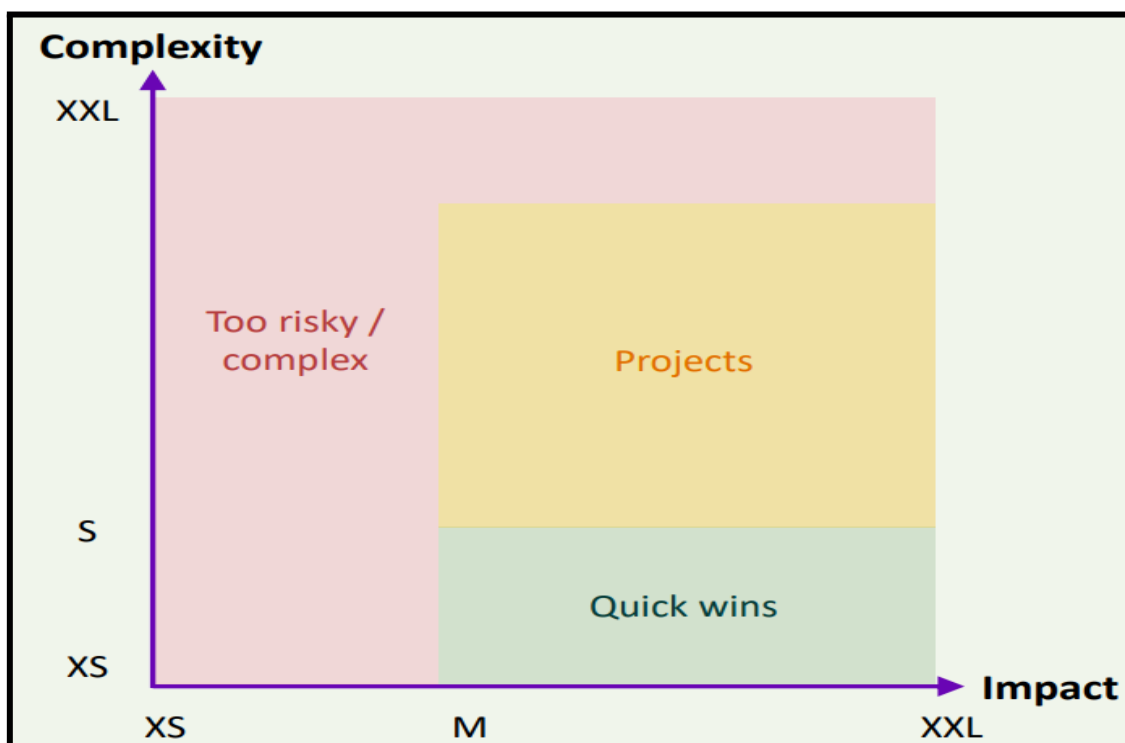
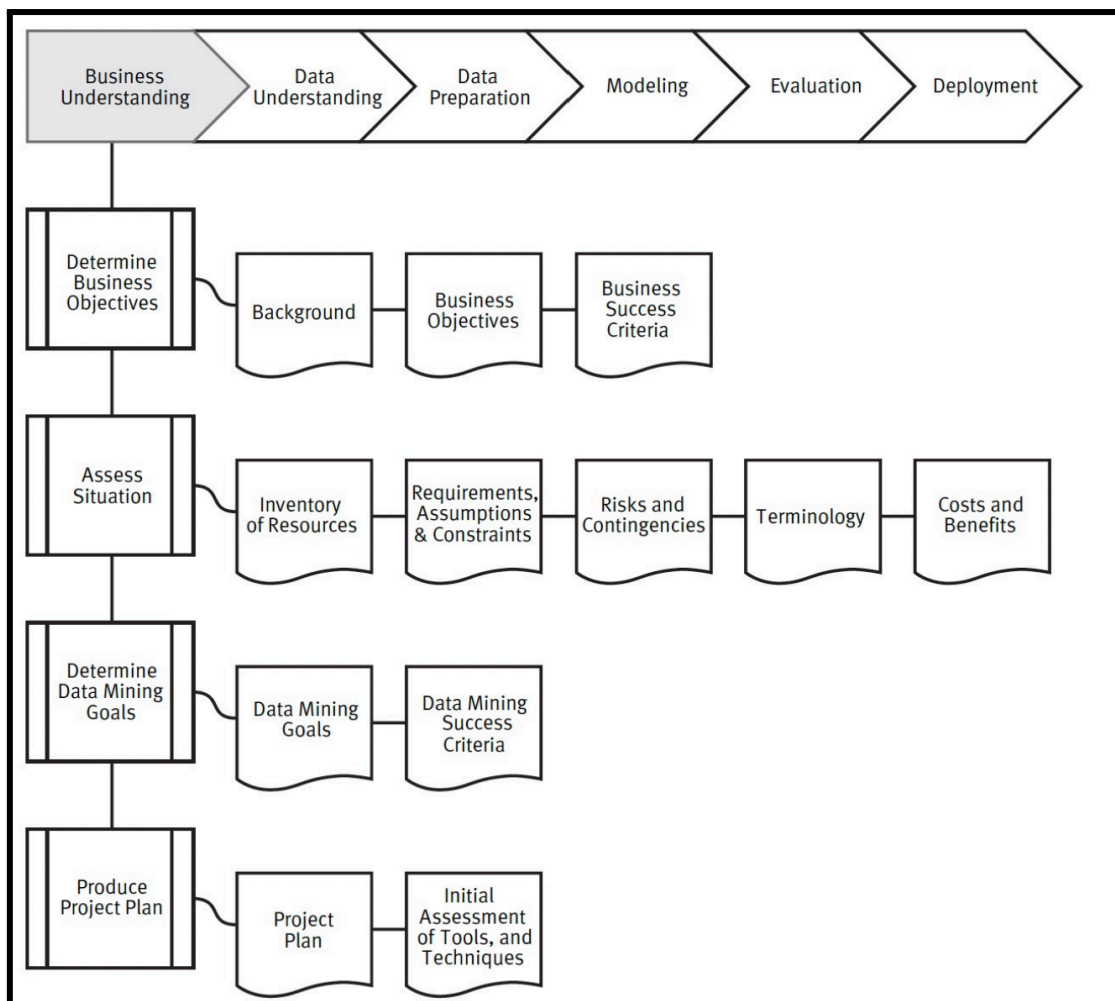
Each model is estimated by number of loops from step 1 to step 6:

- Each loop is 2 weeks long during which a model is deployed
- Once deployment is done a pipeline of production assessment and feedback are derived
- This last are used for the next loop in order to enhance the model

- The time needed for a successful implementation of an end to end solution ready for deployment is estimated by number of iterations, ex: model one takes 4 loops
- **Benefits:**
 - Stakeholders gets a deployed model After just 15 days, while in other methodologies it can take up to 6 months
 - The estimation of the needed duration is more accurate as the only factors to take in consideration are realized in feasibility study
 - A strong collaboration between the domain experts and the AI experts making sure to develop the best AI solutions

Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives Background Business Objectives Business Success Criteria Assess Situation Inventory of Resources Requirements, Assumptions, and Constraints Risks and Contingencies Terminology Costs and Benefits Determine Data Mining Goals Data Mining Goals Data Mining Success Criteria Produce Project Plan Project Plan Initial Assessment of Tools and Techniques	Collect Initial Data Initial Data Collection Report Describe Data Data Description Report Explore Data Data Exploration Report Verify Data Quality Data Quality Report	Select Data Rationale for Inclusion/Exclusion Clean Data Data Cleaning Report Construct Data Derived Attributes Generated Records Integrate Data Merged Data Format Data Reformatted Data Dataset Dataset Description	Select Modeling Techniques Modeling Technique Modeling Assumptions Generate Test Design Test Design Build Model Parameter Settings Models Model Descriptions Assess Model Model Assessment Revised Parameter Settings	Evaluate Results Assessment of Data Mining Results w.r.t. Business Success Criteria Approved Models Review Process Review of Process Determine Next Steps List of Possible Actions Decision	Plan Deployment Deployment Plan Plan Monitoring and Maintenance Monitoring and Maintenance Plan Produce Final Report Final Report Final Presentation Review Project Experience Documentation

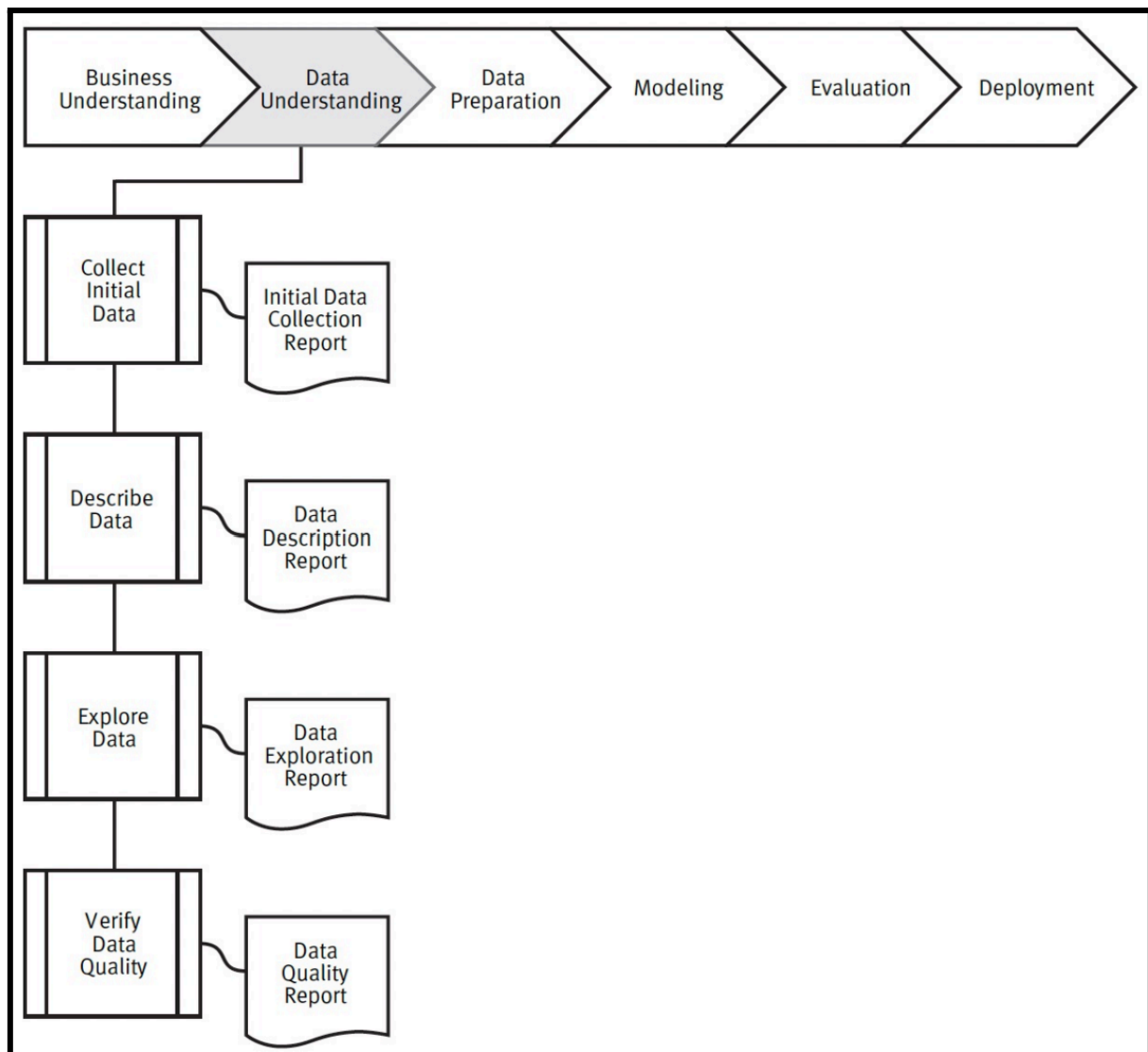
Business understanding:



Risk analysis What can go wrong with a project? • Risk: description • Impact: what are the consequences? • Probability: is it unlikely or not • Mitigation: how to avoid it (or at least limit the effect)

Risk	Potential impact	Probability	Mitigation
Not enough data	Model not powerful enough	Average	Launch the project once 1000 data have been obtained Analyze errors and validate the model only if objectives are met
Unbalanced data	Minority bias	High	Analyze the dataset beforehand to validate balancing Use balancing techniques if necessary
Incomplete data	Errors on rarer cases	Average	Validate dataset coverage Provide the user with feedback on decisions made

Data understanding:



Data source:

It is necessary to indicate where the data come from and to validate the absence of certain biases:

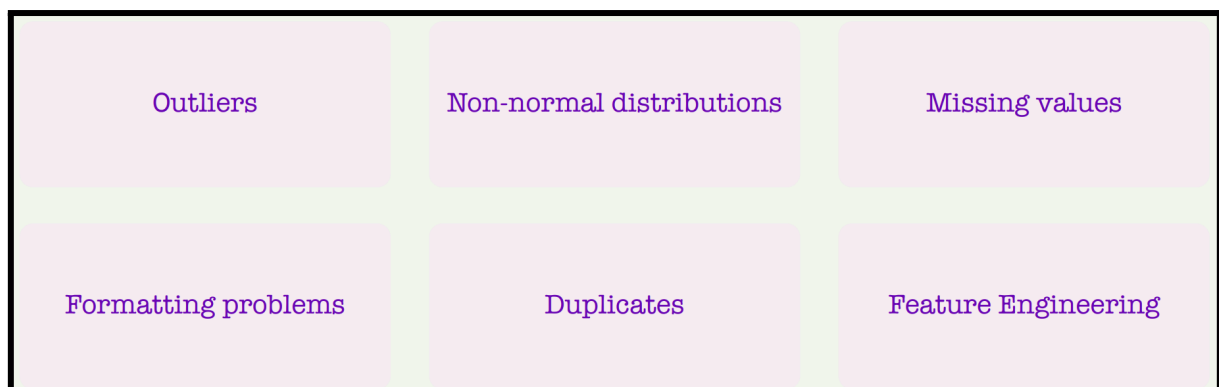
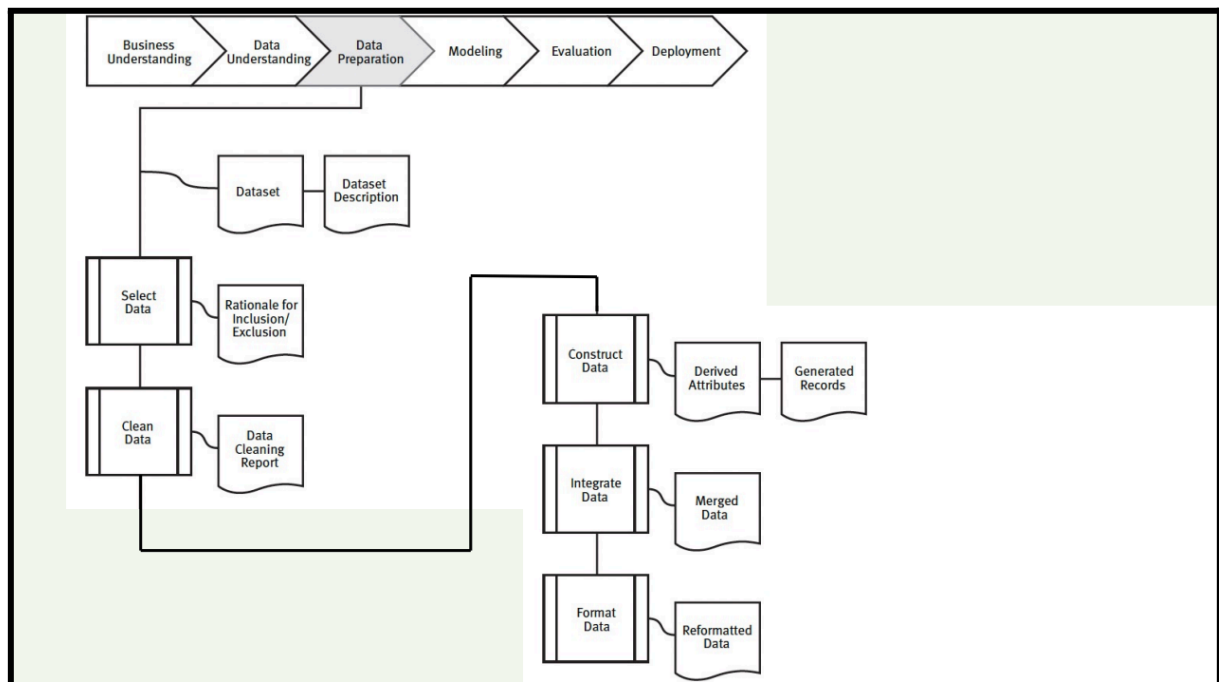
- Missing features
- Data selection during extraction
- Data reliability
- Obsolescence
- Correspondence with field reality

Describe Data:

Data dictionary For each field, you must indicate : • Name • Description • Format or list of possible values and unit • A comment if required In particular, look for potential biases in data definition, in limited choices of values, etc

Explore Data:

Statistical data analysis Two types of statistical analysis : • Univariate statistics: each variable is analyzed individually • Bivariate statistics: each variable is analyzed in relation to a second (usually the target variable)

Verify Data quality**Data preparation:**



Use case example for business understanding and data understanding:

Perform the **Business Understanding + Data Understanding** step for the following dataset:
<https://www.kaggle.com/datasets/rikdifos/credit-cardapproval-prediction>

Business Understanding

Remember to draw up a risk analysis table with the 4 columns: **risk - impact - probability - mitigation**.

For example:

- **Too little data** → model not accurate / biased → high probability → make sure you have enough data before launching the project, retrieve new data as you go along, think about testing on enough data...

These criteria can be:

- **Business-related** (not enough people's time, difficulty in defining the project, little decision-making, etc.),
- **Technical** (difficulty in retrieving data, compatibility problems, etc.),
- **Data science** (not enough data, biased data, etc.).

For **legal analysis**, the main texts are the **RGPD** (personal data) and the **AI Act** (for AI). Additional regulations may apply depending on the field: health data, banking secrecy, etc.

When defining objectives, make sure you separate **business objectives** from **data science objectives**.

- For example, **increasing sales** is a business objective, so is **getting more clicks**.
- On the other hand, achieving **85% accuracy** is a **data science objective**.

Finally, for the **lexicon**, it's important to define terms. In this exercise, it wasn't very clear what **"bad payer"** meant, so it's important to define it more precisely. This is particularly important for the **target variable**.

Data Understanding

The aim of this phase is to **detect problems for the future**.

The first point is to analyze the **data source**. Here, there are two files, i.e. two sources:

1. **Data filled in during card applications** → These are **declarative forms**, so there may be errors, as well as a **significant bias**.
2. **Bank data** → Limited to **customers who have been accepted**. As a result, there is a **major bias** in the usable data.

It would therefore be necessary to have **additional data** from people who have been considered at risk, and a **verification of form data**.

For **statistical analysis**, think carefully about **univariate and bivariate analysis**.

- For **univariate analysis**, you need to look at each field to detect problems:
 - Too many identical values
 - Too many missing values
 - Non-uniform distribution (or Gaussian for numerical variables)
 - Inconsistent fields...

In the example, there were clearly **more women than men**.

There are also problems in the distribution of **children, salary, etc**.

All participants have a **cell phone**, perhaps because the form is filled in via a **mobile application**.

Many modifications will therefore be necessary to the data.

Generic Tips

In all cases, a **good slide** should contain:

- **A text:** This is the message. For example, don't put "**gender distribution**," but "**gender imbalance can lead to bias**," which indicates the message to be retained.
- **A graphic:** Make it as simple as possible, with as little text/visual elements as possible, to retain only the message (in this case, it could be a pink/blue pie chart with $\frac{2}{3}$ - $\frac{1}{3}$).



Example:

<https://www.afponline.org/training-resources/resources/articles/Details/before-and-after-data-visualizations>

Modeling, Evaluation and Deployment are steps to be studied and discussed after the proof of concept

Notes:

Biases do not come from data scientists but from datasets

- **Take the time to define your KPIs:**

- Every KPI is a metric, but not every metric is a KPI.
- KPIs are decision aids, they must be:
 - not numerous (otherwise they may generate noise)
 - aligned with corporate strategy
 - Actionable

- **Mind the gap between expectations and reality**

- Beware the magic side of AI and its deceptions(e.g AI performance versus a deterministic approach)
- AI is the stuff of dreams... but it is also irrationally frightening, which can hinder its use in critical operational conditions.
- Explain as transparently as possible what AI can and cannot do.

- **Never underestimate the cost of integration into an existing ecosystem**

- Create API or specific connector to :
 - Integrate input data
 - Export results (data lake, external dashboard, Alert system, etc.)
 - e.g SCADA (Factories), MQTT (IoT)

- **Pay attention to compatibility with unusual programming languages or frameworks (C++, C#, etc.), Python is still not widely used outside MLops.**

- **Focus on Human Machine Interface (HMI)**

- B2C : use UX/UI best practices (user research, prototype, A/B Test). Start lean.
- B2B : understand acceptance conditions for expert users (e.g. too much notifications/alerts will favour the disconnection of the system)

Case study:

Elevator Pitch:

In a world where health and nutrition have become paramount to a balanced life, **WAWEE** aims to simplify the tracking of individuals' daily diets.

WAWEE is a mobile app that uses AI to identify the foods in your plate and provide accurate nutritional analysis.

Personas :

- People who want to lose weight
- Athletes looking to maximize their performance
- People with food allergies
- People looking for a healthier and balanced diet

Requirements

- The user must download the Wawee app on their mobile device.

The user must have a personal account and has access to their smartphone's camera.

Workflow

1- Food Identification:

The user opens the Wawee app and selects the option to take a picture of their plate.

The AI analyzes the image and identifies the foods present.

The user receives a list of detected foods and can edit or confirm the entries.

2 - Nutritional analysis:

Once the foods are confirmed, the app calculates the total calories as well as the distribution of macronutrients: fat, carbohydrates and proteins.

The user can see a nutritional summary for each food as well as for the complete meal.

3 - Nutritional Suggestions:

Based on the user's nutritional information and health goals, the app suggests improvements for future meals.

The user can record the meal in their food diary or schedule suggested meals for the following days.

Expected results:

The user has a detailed account of the nutritional values of his meal.

The user has suggestions to improve their diet according to their personal goals.

Success Criteria:

The app must accurately recognize a wide variety of foods.

The app must provide nutritional information that is in line with current scientific data.

Food suggestions should be customized according to the user's preferences and goals.

Constraints :

The app must maintain the privacy and security of the user's data.

The app should be intuitive and easy to use, minimizing the need for manual intervention.

The app should work effectively on iOS & Android.

Purpose of the work:

1) Define KPIs

2) Define development plan by phase (you can use the framework of your choice)

3) Define a target organization for production teams

4) Define a product roadmap

Solution:**1. Define KPIs*****User Engagement:***

Daily Active Users (DAU): Number of users actively using the app daily.

Weekly Active Users (WAU): Number of users actively using the app weekly.

Retention Rate: Percentage of users who continue using the app after a specified time (e.g., 1 week, 1 month).

Feature Usage:

Food Recognition Accuracy: Percentage of foods correctly identified by the AI compared to manual verification.

Nutritional Data Accuracy: Alignment of the app's nutritional information with established scientific data.

Personalized Suggestions Adoption: Percentage of users who adopt or interact with personalized dietary suggestions.

User Satisfaction:

Net Promoter Score (NPS): User satisfaction rating to gauge how likely users are to recommend the app to others.

App Store Ratings & Reviews: Average rating and the nature of reviews on the iOS and Android app stores.

Business Growth:

User Growth Rate: Month-over-month growth in the number of new users.

Churn Rate: Percentage of users who stop using the app over time.

Conversion Rate: Percentage of free users upgrading to premium features (if any).

Technical Performance:

Crash-Free Sessions: Percentage of user sessions without any app crashes.

Load Time: Time taken for the app to load, ensuring it's under 2 seconds.

Privacy & Security Compliance: Number of privacy or security issues reported and resolved.

• 2. Define Development Plan by Phase

- To organize the development process, WE will use the Standard AI product development framework:

- Feasibility study

- **Objectives:**

- Define the product vision and goals.
 - Identify user personas and their pain points.
 - Gather technical and market requirements.
 - Conduct a feasibility study.
 - Define the MVP (Minimum Viable Product) scope.

- **Key Activities:**

- Workshop sessions with stakeholders.
 - Market research and competitive analysis.
 - Drafting user stories and product backlog.
 - Initial technology stack selection.
 - Identifying key external APIs or data sources. • Proof of concept

- **Objectives:**

- Validate core concepts and user flows through prototypes.
 - Develop user experience (UX) and user interface (UI) designs.

- **Key Activities:**

- Wireframing key user journeys.
 - Designing high-fidelity mockups.
 - Creating interactive prototypes for user testing.
 - Conducting usability tests with potential users.
 - Iterating based on user feedback.



- Pilot
- **Objectives:**
 - Build the core features of the app as defined in the MVP.
 - Ensure cross-platform compatibility (iOS & Android).
- **Key Activities:**
 - Setting up the development environment.
 - Developing key features: account creation, food recognition, nutritional analysis, personalized suggestions.
 - Integration with external APIs for nutritional data.
 - Implementing security and privacy measures.
 - Conducting unit tests and continuous integration.
- Product
- **Objectives:**
 - Release the app to the public and monitor its performance.
 - Ensure continuous improvement and feature updates.
- **Key Activities:**
 - Launching the app on iOS and Android stores.
 - Monitoring KPIs (user engagement, technical performance, etc.).
 - Implementing a feedback loop with users for continuous improvement.
 - Rolling out updates and new features based on user demand and feedback.

• 3. Define a Target Organization for Production Teams

- Given the early-stage nature of the startup, a lean and agile team structure will be ideal. The structure should be flexible, allowing for quick iteration and scaling as the product and company grow.
- **Core Teams:**
- **Product Management:**
 - **Product Manager (You):** Responsible for overall product vision, roadmap, and alignment with business goals. Work closely with all teams to ensure the product meets user needs and success criteria.
- **Engineering:**
 - **CTO (Young Engineer):** Lead the technical team, set the technical vision, and ensure high code quality. Since they lack experience, they will also need mentorship or external consultancy.
- **Mobile Developers (2-3):** Focus on iOS and Android development.
- **Backend Developer (1-2):** Handle server-side logic, database management, and integration with external APIs.



- **AI Specialist (1):** Develop and refine the food recognition algorithms.
- **Design & User Experience:**
- **UI/UX Designer (1):** Design the app's interface and ensure a seamless user experience.
- **Quality Assurance (QA):**
- **QA Engineer (1):** Conduct manual and automated testing to ensure product quality.
- **Operations & Support:**
- **Customer Support (1):** Handle user inquiries, feedback, and support issues post-launch.
- **Advisory Roles:**
- **Product Advisor (Part-time):** An experienced product leader to guide product strategy and mentor the product manager and CTO.
- **Legal/Compliance Advisor (Part-time):** Ensure the app meets all legal requirements, especially around data privacy.
- 4. Define a Product Roadmap
- A clear and strategic roadmap is crucial to keeping the team aligned and focused. Below is a high-level product roadmap.
- **Q1: Discovery & Prototyping**
- Finalize product vision and objectives.
- Complete market research and persona development.
- Develop initial wireframes and prototypes.
- Conduct initial user testing and iterate on design.
- **Q2: MVP Development**
- Begin development of core features (account creation, food recognition, nutritional analysis).
- Implement cross-platform capabilities for iOS and Android.
- Set up backend infrastructure and integrate necessary APIs.
- Perform internal alpha testing and debugging.
- **Q3: Beta Testing & Refinement**
- Launch beta version to a select group of users.
- Gather feedback and identify key improvements.
- Refine AI algorithms for better food recognition accuracy.



- Ensure compliance with privacy and security regulations.
- ***Q4: Public Launch & Growth***
- Officially launch WAWEE on the iOS and Android app stores.
- Implement marketing strategies to drive user acquisition.
- Monitor key KPIs and address any post-launch issues.
- Begin work on additional features (e.g., advanced meal planning, social sharing).
- ***Q1 Next Year: Feature Expansion***
- Introduce premium features and explore monetization strategies.
- Roll out additional dietary support (e.g., for specific diets like keto or vegan).
- Expand AI capabilities to cover a wider variety of foods.
- Scale customer support and infrastructure to handle increased user base.
- This plan and organization should provide a solid foundation for WAWEE to achieve its goals and successfully deliver its first version to production.

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